Preface

To the Reader

The book you have in your hands is the second volume of the twelfth edition of *Core Java*, fully updated for Java 17. The first volume covers the essential features of the language; this volume deals with the advanced topics that a programmer needs to know for professional software development. Thus, as with the first volume and the previous editions of this book, we are still targeting programmers who want to put Java technology to work in real projects.

As is the case with any book, errors and inaccuracies are inevitable. Should you find any in this book, we would very much like to hear about them. Of course, we would prefer to hear about them only once. For this reason, we have put up a web site at **http://horstmann.com/corejava** with a FAQ, bug fixes, and workarounds. Strategically placed at the end of the bug report web page (to encourage you to read the previous reports) is a form that you can use to report bugs or problems and to send suggestions for improvements for future editions.

About This Book

The chapters in this book are, for the most part, independent of each other. You should be able to delve into whatever topic interests you the most and read the chapters in any order.

In **Chapter 1**, you will learn all about the Java stream library that brings a modern flavor to processing data, by specifying what you want without describing in detail how the result should be obtained. This allows the stream library to focus on an optimal evaluation strategy, which is particularly advantageous for optimizing concurrent computations.

The topic of **Chapter 2** is input and output handling (I/O). In Java, all input and output is handled through input/output streams. These streams (not to be confused with those in **Chapter 1**) let you deal, in a uniform manner, with communications among various sources of data, such as files, network connections, or memory blocks. We include de-

tailed coverage of the reader and writer classes that make it easy to deal with Unicode. We show you what goes on under the hood when you use the object serialization mechanism, which makes saving and loading objects easy and convenient. We then move on to regular expressions and working with files and paths. Throughout this chapter, you will find welcome enhancements in recent Java versions.

Chapter 3 covers XML. We show you how to parse XML files, how to generate XML, and how to use XSL transformations. As a useful example, we show you how to specify the layout of a Swing form in XML. We also discuss the XPath API, which makes finding needles in XML haystacks much easier.

Chapter 4 covers the networking API. Java makes it phenomenally easy to do complex network programming. We show you how to make network connections to servers, how to implement your own servers, and how to make HTTP connections. This chapter includes coverage of the new HTTP client.

Chapter 5 covers database programming. The focus is on JDBC, the Java database connectivity API that lets Java programs connect to relational databases. We show you how to write useful programs to handle realistic database chores, using a core subset of the JDBC API. (A complete treatment of the JDBC API would require a book almost as big as this one.)

Java had two prior attempts at libraries for handling date and time. The third one was the charm in Java 8. In **Chapter 6**, you will learn how to deal with the complexities of calendars and time zones, using the new date and time library.

Chapter 7 discusses a feature that we believe can only grow in importance: internationalization. The Java programming language is one of the few languages designed from the start to handle Unicode, but the internationalization support on the Java platform goes much further. As a result, you can internationalize Java applications so that they cross not only platforms but country boundaries as well. For example, we show you how to write a retirement calculator that uses either English, German, or Chinese languages.

Chapter 8 discusses three techniques for processing code. The scripting and compiler APIs allow your program to call code in scripting languages such as JavaScript or Groovy, and to compile Java code. Annotations allow you to add arbitrary information (sometimes called metadata) to a Java program. We show you how annotation processors can harvest these annotations at the source or class file level, and how annotations can be used to influence the behavior of classes at runtime. Annotations are only useful with

tools, and we hope that our discussion will help you select useful annotation processing tools for your needs.

In **Chapter 9**, you will learn about the Java Platform Module System that was introduced in Java 9 to facilitate an orderly evolution of the Java platform and core libraries. This module system provides encapsulation for packages and a mechanism for describing module requirements. You will learn the properties of modules so that you can decide whether to use them in your own applications. Even if you decide not to, you need to know the new rules so that you can interact with the Java platform and other modularized libraries.

Chapter 10 takes up the Java security model, user authentication, and the cryptographic functions in the Java security library. You will learn about important features such as message and code signing, authorization and authentication, and encryption. We conclude with examples that use the AES and RSA encryption algorithms.

Chapter 11 contains all the Swing material that didn't make it into Volume I, especially the important but complex tree and table components. We also cover the Java 2D API, which you can use to create realistic drawings and special effects. Of course, not many programmers need to program Swing user interfaces these days, so we pay particular attention to features that are useful for images that can be generated on a server.

Chapter 12 takes up native methods, which let you call methods written for a specific machine such as the Microsoft Windows API. Obviously, this feature is controversial: Use native methods, and the cross-platform nature of Java vanishes. Nonetheless, every serious programmer writing Java applications for specific platforms needs to know these techniques. At times, you need to turn to the operating system's API for your target platform when you interact with a device or service that is not supported by Java. We illustrate this by showing you how to access the registry API in Windows from a Java program.

As always, all chapters have been completely revised for the latest version of Java. Outdated material has been removed, and the new APIs up to Java 17 are covered in detail.

Conventions

As is common in many computer books, we use `monospace type` to represent computer code.



Note

Notes are tagged with “note” icons that look like this.



Tip

Tips are tagged with “tip” icons that look like this.



Caution

When there is danger ahead, we warn you with a "caution" icon.



C++ Note

There are a number of C++ notes that explain the difference between the Java programming language and C++. You can skip them if you aren't interested in C++.

Java comes with a large programming library, or Application Programming Interface (API). When using an API call for the first time, we add a short summary description at the end of the section. These descriptions are a bit more informal but, we hope, also a little more informative than those in the official online API documentation. The names of interfaces are in italics, just like in the official documentation. The number after a class, interface, or method name is the JDK version in which the feature was introduced.

Application Programming Interface 1.2

Programs whose source code is included in the companion code for this book are listed as examples, for instance

Listing 1.1 ScriptTest.java

You can download the companion code from **http://horstmann.com/corejava**.

Register your copy of *Core Java, Volume II: Advanced Features, Twelfth Edition*, on the InformIT site for convenient access to updates and/or corrections as they become available. To start the registration process, go to **informit.com/register** and log in or create an account. Enter the product ISBN (9780137871070) and click Submit. Look on the Registered Products tab for an Access Bonus Content link next to this product, and follow that link to access any available bonus materials. If you would like to be notified of exclusive offers on new editions and updates, please check the box to receive email from us.

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